

The Symmetric Banach-Sphere Entropy Estimate Extends To The Endpoint Window

Automatic math-discovery smoke test

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Source Problem

The source paper is Aicke Hinrichs and Sebastian Mayer, *Entropy Numbers of Spheres in Banach and quasi-Banach spaces*, arXiv:1501.03680; later published in *Journal of Approximation Theory* 200 (2015), 144–152, DOI: 10.1016/j.jat.2015.05.004 [1]. The relevant signal is Remark 14 on page 9 of the arXiv PDF, immediately after Corollary 13.

Remark 14. In contrast to Theorem 7, the statement of Corollary 13 cannot be extended to $k \geq d$ in a simple way as we do not know how $e_d(\mathbb{S}_X^{d-1}, Y)$ behaves.

In text form, the remark says that, unlike the concrete ℓ_p -sphere result in Theorem 7, Corollary 13 is not extended to $k \geq d$ because the authors do not know how $e_d(\mathbb{S}_X^{d-1}, Y)$ behaves.

Result Type

This packet gives a **candidate full endpoint extension** of Corollary 13 for the symmetric Banach-space setting considered in the remark.

Definitions And Source Inputs

Entropy numbers are dyadic: $e_k(A, Y)$ is the infimum of radii $\varepsilon > 0$ such that A is covered by 2^{k-1} translates of εB_Y .

Let X and Y be d -dimensional Banach spaces on $\mathbb{R}^d$, with normalized canonical bases, whose norms are symmetric with respect to permutations and sign changes of the coordinates. Write

$$\mathbb{S}_X = \{x \in \mathbb{R}^d : \|x\|_X = 1\}$$

and

$$\lambda_E(m) = \left\| \sum_{i=1}^m e_i \right\|_E \quad (1 \leq m \leq d)$$

for a symmetric d -dimensional space E .

We use two source inputs from Hinrichs–Mayer [1]. First, their Theorem 9 gives, for the coordinate hyperplanes H_i ,

$$e_k(\mathbb{S}_X, Y) \geq \max_i e_k(B_{X^i}, Y^i).$$

Second, their proof of Corollary 13 invokes Schuett's entropy estimates for balls in symmetric Banach spaces [2, Theorem 5]. In the form needed here, if E, F are n -dimensional symmetric Banach spaces with normalized bases, then for $k \geq n$

$$e_k(B_E, F) \geq \frac{1}{e} 2^{-k/n} \frac{\lambda_F(n)}{\lambda_E(n)}$$

and, with an absolute constant c_S ,

$$e_k(B_E, \ell_\infty^n) \leq c_S 2^{-k/n} \frac{1}{\lambda_E(n)}.$$

These are exactly the lower and upper estimates used in equations (2) and (3) of the source proof, with the dimension changed as appropriate.

Proof Intuition

The source proof already contains the hard lower bound: project the sphere onto a coordinate hyperplane and apply Schuett's lower estimate for the $(d-1)$ -dimensional ball. This works for all $k \geq d$, not just the large k range printed in Corollary 13.

For the upper bound, the only missing interval is finite:

$$d \leq k < 2d + \lceil \log d \rceil - 1.$$

In that interval monotonicity reduces everything to $e_d(\mathbb{S}_X, Y)$, and $\mathbb{S}_X \subset B_X$. The whole ball B_X can be covered in the ℓ_∞^d metric at entropy level d by Schuett's ball estimate, and then transported into Y through the identity map $\ell_\infty^d \rightarrow Y$, whose norm is $\lambda_Y(d)$. Symmetry and normalization compare $\lambda_Y(d)$ and $\lambda_X(d)$ back to the $(d-1)$ -level quantities in Corollary 13. The factor $2^{-k/(d-1)}$ is bounded below by an absolute constant throughout this finite endpoint window, so the same asymptotic form follows.

Main Result

Theorem 1. *Let $d \geq 2$, and let X and Y be symmetric Banach spaces on $\mathbb{R}^d$ with normalized canonical bases. Then there is an absolute constant $C > 0$ such that, for every $k \geq d$,*

$$\frac{1}{e} 2^{-k/(d-1)} \frac{\lambda_Y(d-1)}{\lambda_X(d-1)} \leq e_k(\mathbb{S}_X, Y) \leq C 2^{-k/(d-1)} \frac{\lambda_Y(d-1)}{\lambda_X(d-1)}.$$

Consequently, the two-sided estimate of Corollary 13 in the source paper extends from $k \geq 2d + \lceil \log d \rceil - 1$ down to all $k \geq d$.

Proof. We first record two elementary consequences of symmetry. The canonical basis is 1-unconditional. Indeed, a coordinate projection is obtained by averaging sign changes over the deleted coordinates, so convexity of the norm gives $\|P_A x\|_E \leq \|x\|_E$ for every coordinate subset A . Hence $\lambda_E(m)$ is nondecreasing in m , and $\lambda_E(m) \geq 1$ for $m \geq 1$. Also,

$$\lambda_E(d) \leq \lambda_E(d-1) + \lambda_E(1) \leq 2\lambda_E(d-1).$$

Finally,

$$\left\| \text{id} : \ell_\infty^d \rightarrow E \right\| = \lambda_E(d),$$

because the maximum of a convex symmetric norm on the cube $[-1, 1]^d$ is attained at a sign vector, and every sign vector has E -norm $\lambda_E(d)$.

For the lower bound, fix a coordinate hyperplane, say H_1 . Since the coordinate projection is contractive, $X^1 = X_1$ and $Y^1 = Y_1$ isometrically in the notation of the source paper. The hyperplane spaces X_1, Y_1 are $(d-1)$ -dimensional symmetric Banach spaces, and

$$\lambda_{X_1}(m) = \lambda_X(m), \quad \lambda_{Y_1}(m) = \lambda_Y(m) \quad (1 \leq m \leq d-1).$$

Theorem 9 of the source paper and Schuett's lower estimate in dimension $d-1$ yield, for every $k \geq d$ and therefore for every $k \geq d-1$,

$$e_k(\mathbb{S}_X, Y) \geq e_k(B_{X_1}, Y_1) \geq \frac{1}{e} 2^{-k/(d-1)} \frac{\lambda_Y(d-1)}{\lambda_X(d-1)}.$$

It remains to prove the upper bound. For

$$k \geq 2d + \lceil \log d \rceil - 1,$$

this is precisely the upper estimate already proved in Corollary 13 of the source paper, with constant $32c$.

Now assume

$$d \leq k < 2d + \lceil \log d \rceil - 1.$$

By monotonicity of entropy numbers and by $\mathbb{S}_X \subset B_X$,

$$e_k(\mathbb{S}_X, Y) \leq e_d(\mathbb{S}_X, Y) \leq e_d(B_X, Y).$$

Factor the identity through ℓ_∞^d :

$$e_d(B_X, Y) \leq \left\| \text{id} : \ell_\infty^d \rightarrow Y \right\| e_d(B_X, \ell_\infty^d).$$

By the preceding observation, $\left\| \text{id} : \ell_\infty^d \rightarrow Y \right\| = \lambda_Y(d)$, while Schuett's upper estimate for the d -dimensional ball gives

$$e_d(B_X, \ell_\infty^d) \leq c_S 2^{-1} \frac{1}{\lambda_X(d)} \leq c_S \frac{1}{\lambda_X(d)}.$$

Therefore

$$e_k(\mathbb{S}_X, Y) \leq c_S \frac{\lambda_Y(d)}{\lambda_X(d)} \leq 2c_S \frac{\lambda_Y(d-1)}{\lambda_X(d-1)}.$$

The same elementary estimate used in the proof of Theorem 7 of the source paper gives

$$2^{-k/(d-1)} \geq 2^{-6}$$

throughout this finite window. Hence

$$e_k(\mathbb{S}_X, Y) \leq 128c_S 2^{-k/(d-1)} \frac{\lambda_Y(d-1)}{\lambda_X(d-1)}.$$

Combining this with the already-known large- k upper bound proves the claim, for example with $C = \max\{32c, 128c_S\}$. $\square$

Verification Notes

No numerical computation is used. The proof depends on three checkable points. First, normalized symmetric Banach bases are 1-unconditional, so coordinate projections are contractions. Second, the factorization norm needed in the finite-window estimate is $\left\| \text{id} : \ell_\infty^d \rightarrow Y \right\| = \lambda_Y(d)$. This is the orientation used explicitly in the proof above. Third, Schuett's ball estimate is applied at entropy level d in dimension d , which is inside its $k \geq n$ range.

The source paper has a small notational inconsistency around the factorization orientation in Theorem 12/Corollary 13: the displayed norm is written in one place as $Y \rightarrow \ell_\infty^d$, while the proof line and the use of $\lambda_Y(d)$ correspond to $\ell_\infty^d \rightarrow Y$. The endpoint argument above uses only the latter, direct factorization.

Novelty And Literature Check

Before promotion, the local run indexes

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registry_index.tsv,  solutions/index.tsv,  
attempts/index.tsv,  proof_gaps/index.tsv
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were searched for arXiv:1501.03680, the title, “entropy numbers of spheres”, “symmetric Banach spaces”, and endpoint phrases around $e_d(\mathbb{S}_X, Y)$. No duplicate solution, prior attempt, or proof-gap packet for this endpoint was found.

A bounded web search on July 19, 2026 for exact phrases from Remark 14 and close variants such as “ $e_d(\mathbb{S}_X, Y)$ ”, “Entropy numbers of spheres symmetric Banach spaces”, and the paper title found bibliographic pages for the source article, including its Journal of Approximation Theory DOI page, but no separate later paper explicitly resolving the endpoint extension.

Limitations

This proves the endpoint extension only in the symmetric Banach-space setting of Corollary 13. It does not address possible extensions for arbitrary quasi-Banach spaces or for non-symmetric monotone spaces beyond the hypotheses of that corollary.

Human Review Recommendation

Send for human review as a candidate full endpoint solution. The recommended review focus is the finite-window upper bound

$$e_d(B_X, Y) \lesssim \frac{\lambda_Y(d-1)}{\lambda_X(d-1)}$$

and the exact invocation of Schuett’s Theorem 5(2) at dimension d .

References

- [1] A. Hinrichs and S. Mayer, *Entropy Numbers of Spheres in Banach and quasi-Banach spaces*, arXiv:1501.03680; *Journal of Approximation Theory* 200 (2015), 144–152, DOI: 10.1016/j.jat.2015.05.004.
- [2] C. Schuett, *Entropy numbers of diagonal operators between symmetric Banach spaces*, *Journal of Approximation Theory* 40 (1984), 121–128.